Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/10/2020
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Project Title:	Utilization of the Smoke Aerosol Measurement Experiment Data for Advanced Modeling and Simulation of Smoke Generation in Micro-Gravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Task Description:	Fire produces heat, smoke, and toxic gases that can threaten human life. Fire detection is a key step in mitigating the risk to life and material. Smoke from fires contains particles that vary in size from 10 nanometers to 10 microns in diameter. For example, smoke generated during the smoldering stage of a fire tend to be fairly large – 0.3 to 10 μm – whereas smoke from the flaming stage are much smaller – 0.01 to 0.3 μm. Photo-electric and ionization smoke detectors are designed to detect these combustion particles. However, these fire detection systems were all designed based on smoke generation, or fire dynamics, in the terrestrial environment where gravity generates buoyancy. Without buoyancy, mixing and heat transfer are reduced, the time-temperature history of the combustion vapors (or particle precursors) and particle trajectories are significantly altered, changing the composition of the smoke. The chemical composition of the burning material also affects the characteristics of the smoke generated. For example, Kapton is regularly used in a variety of space applications, from thermal covering in the Ultra Heavy Cosmic Ray Experiment to wiring insulation in spacecraft. Studies have shown that under certain conditions, Kapton ages poorly and is a relatively high risk for fire. An alternative is Pyrell, a fire retardant modified polyester polyurethane foam. While Kapton tends to produce spherical particles when burning, Pyrell tends to produce “fractal-like” aggregates. Spheres, rods, and flakes have different transport properties and their rate of collision and growth are sensitive to particle shape. Recent NASA investigations into smoke generation were performed on the International Space Station (ISS). The Smoke Aerosol Measurement Experiment (SAME) provides aerosol data on the burning of several materials in a spacecraft low gravity environment. The data reveal dynamics and properties different to those in normal gravity. Whereas physical investigation aboard the ISS is prohibitive, numerical simulation can be used to explore the origins of these dynamics or the range of parameters that affect smoke properties. My research group has developed a rich set of modeling tools to predict the dynamics of molecular clusters and nanoparticles in complex flow systems. By better understanding the interaction between buoyancy, heat transfer, and particle formation, transport, and growth, we can create advanced modeling and simulation tools that enable improved design of future fire detectors in space systems. This proposal contains a comprehensive plan to develop models that exhibit a high degree of fidelity to the underlying physics and chemistry and simulate low gravity smoke generation. The SAME data are valuable to aerosol scientists as it can be used to generate new physical insights into polymer particle formation. “Traditional” fires yield a variety of combustion products – hydrocarbon soot, condensed water vapor, ash particles, as well as unburned and/or re-condensed original materials. Analyzing the aerosol properties of such a mixture is quite involved and challenging. However, the SAME investigation aims to shed light on the smoldering stages of a fire – when the number and complexity of the interactions are significantly reduced – e.g., pyrolysis, nucleation, condensation, coagulation, and particle transport. In addition to improving fire detection in microgravity, mission protection and our ability to develop better safeguards for space exploration, the SAME provides a comprehensive database for use in the development and benchmarking of aerosol formation and growth models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing models and tools that characterize particles generated as smoke. We are using these models and tools to ascertain how temperature, gravity, and material properties affect the smoke produced. This would directly lead to better smoke detection devices and prevent fires that can lead to loss of materiel and loss of life. The tools can be run on desktop workstations facilitating engineering use.
Task Progress:	This work examines how microgravity conditions on spacecraft change the development of smoke aerosol relative to smoke on Earth. We perform simulations of particle formation and growth with the goal of capturing smoke dynamics similar to those realized in experiments performed aboard the International Space Station (ISS). The simulations provide detailed information about fluid/thermal/particle interactions that cannot be acquired experimentally on orbit. A surrogate material, dibutyl pthalate, is expelled from a hot surface in a rectangular chamber while ambient air is blown through the domain, convecting the developing smoke downstream. The study considers two flow rates, two geometric configurations, and the isolated effects of gravitational particle settling and thermophoresis. In Earth gravity, buoyancy produces asymmetries in all fields of interest and amplifies vapor-to-particle conversion near room temperature surfaces at the upper boundary of the simulation domain. Gravitational settling removes large particles from the aerosol, limiting the maximum particle size to ~40 μm and the mode particle size to ~5 μm. Without buoyancy or settling, the microgravity smoke plume is spatially symmetric and contains very large particles (> 40 μm). A bimodal particle size distribution results in microgravity when prominent recirculation occurs behind the hot vapor source. The distribution is comprised of large particles (> 20 μm) associated with long residency time, vapor rich, high temperature recirculation, and smaller particles (< 10 μm) that originate farther downstream of the hot vapor source. A distinct large particle mode does not appear in microgravity flows that do not have prominent high temperature recirculation, although the aerosol still contains large particles. Without recirculation, the mode particle diameter in microgravity is ~10 μm, or approximately twice that of terrestrial smoke. Results show that future experiments should carefully consider and quantify surface interactions and the presence of hot, recirculating vapor, in order to control geometry-dominated features of the particle size distribution. A manuscript based on this work will be submitted to the journal Aerosol Science & Technology. The final document, when accepted, will be provided to NASA. All research and simulations have been completed. We were able to simulate/replicate the Smoke Aerosol Measurement Experiment (SAME) performed aboard the ISS.
Bibliography Type:	Description: (Last Updated:)